

21-06-2026

## Agenda:

ML-10

- Logistic Regression practical
- ROC curve
- cross-validation split
- KNN

## sklearn pipeline & column transformer

Task a → call google.com & search → KNA def task\_a()  
Task b → check the output of 10 using sigmoid def task\_b()  
Task c → use task b as input & predict  $\hat{y}$  def task\_c()  
Task d → join zoom meeting def task\_d()

task b() → task c()

task a()

task d()

pipeline ( ("1", b),  
("2", c) )

sequence

columnTrn ( ("3", a),  
("4", b) )

parallel

transformer ( (pipeline), (columnTrn) )  
↓  
fit

→ Q: If we want to provide our custom threshold, then what will it be?

$\rightarrow 0.5$   
 $\rightarrow 0.6$   
 $\rightarrow 0.9$

$> 0.5 \rightarrow 1$   
 $< \rightarrow 0$

ROC curve :-

TP: model predict positive, actual positive

TN: model predicts negative, actual negative

FP: model predict positive, actual negative

FN: model predicts negative, actual positive

TPR → How many actual positive we caught correctly

$$= \frac{TP}{TP + FN}$$

Recall

FPR → how many negative were wrongly marked as positive

$$= \frac{FP}{FP + TN}$$

| Email | y | $\hat{y}$ | Threshold(0.5) |
|-------|---|-----------|----------------|
| A     | 1 | 0.95      | 1              |
| B     | 0 | 0.7       | 0              |
| C     | 1 | 0.6       | 0              |
| D     | 0 | 0.4       | 0              |
| E     | 1 | 0.2       | 0              |

$\rightarrow TP \rightarrow 1 (A)$      $TN \rightarrow 2 (B, D)$   
 $FP \rightarrow 0$      $FN \rightarrow 2 (C, E)$

$TPR \rightarrow TP / (TP + FN)$   
 $= 1 / 3 = 0.33$

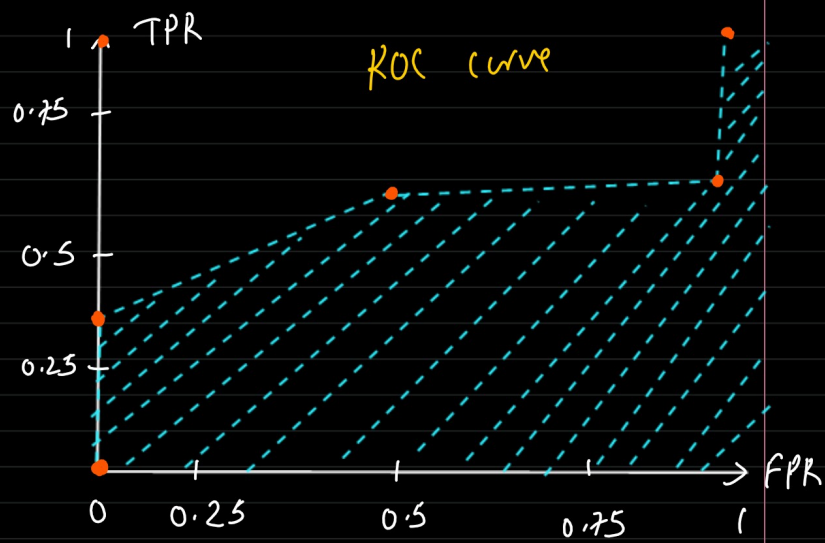
Actual spam → A, C, E

not-spam → B, D

$$FPR = \frac{FP}{FP + TN}$$

$$= 0 / (0 + 2) = 0$$

| Threshold | TPR  | FPR |
|-----------|------|-----|
| 0         | 1.0  | 1.0 |
| 0.4       | 0.67 | 1.0 |
| 0.6       | 0.67 | 0.5 |
| 0.8       | 0.33 | 0   |
| 1.0       | 0    | 0   |



Model-1

ROC  $\rightarrow$  TPR vs FPR for all threshold

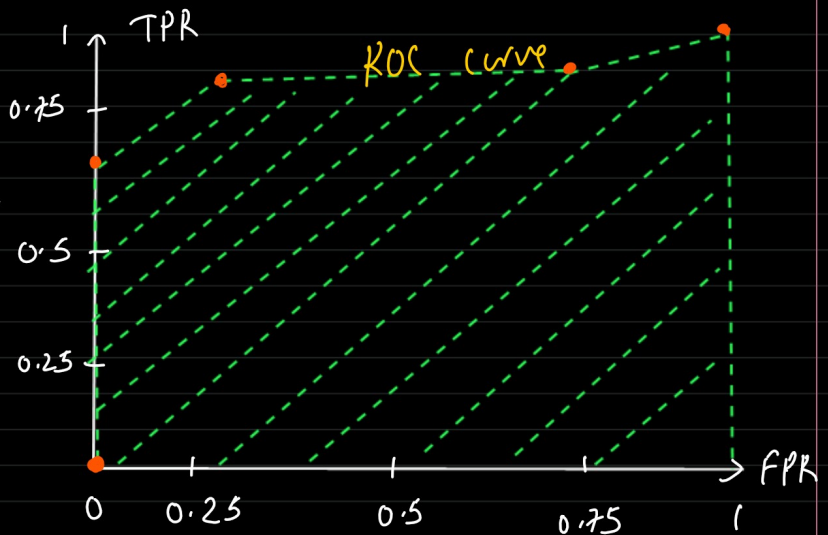
FPR  $\rightarrow$  0

TPR  $>$  FPR  $\rightarrow$  good

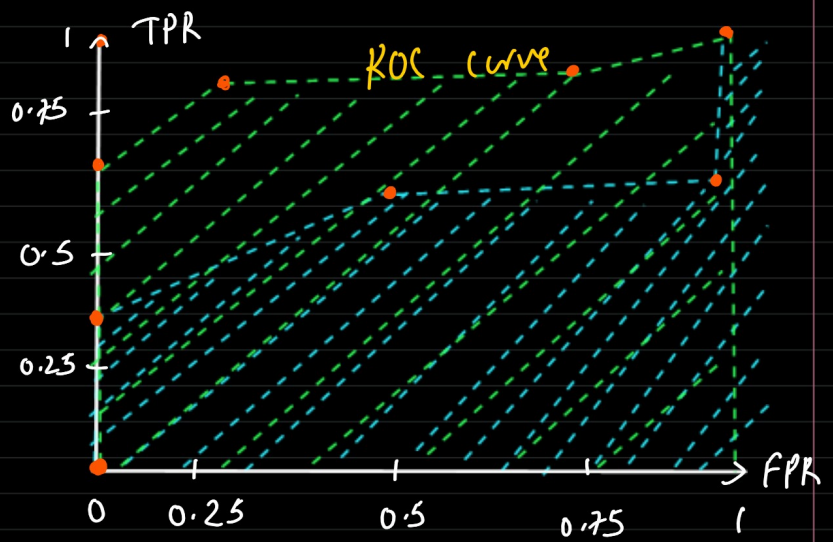
TPR  $\rightarrow$  1

AUC  $\rightarrow$  Area under curve

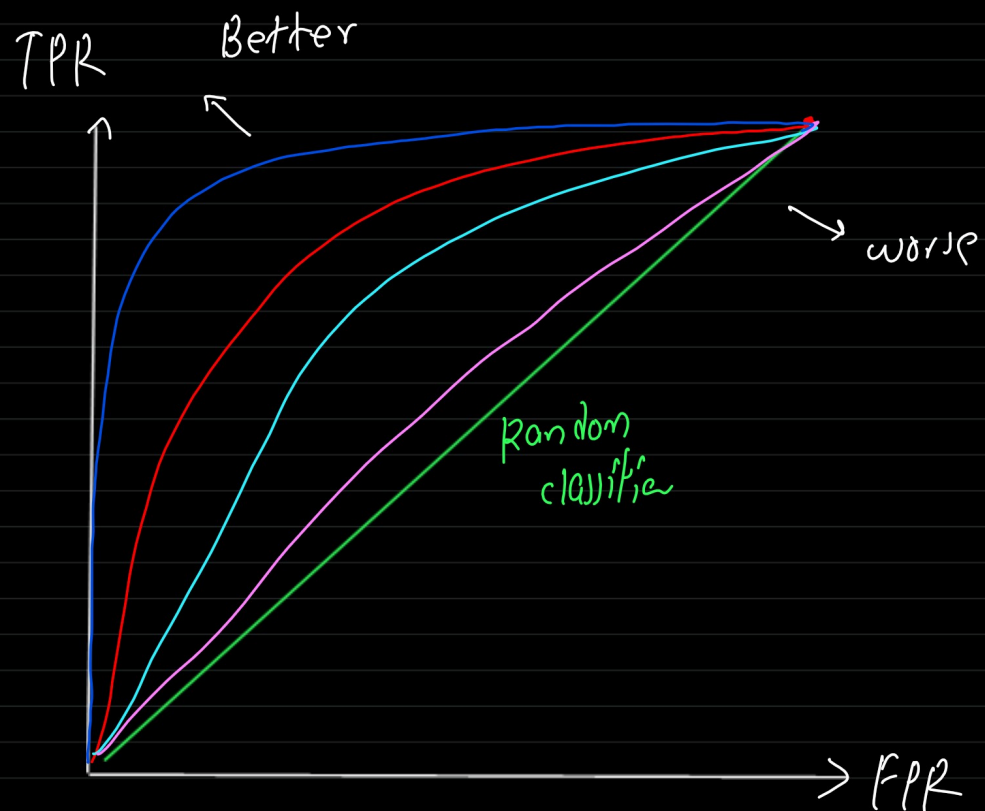
$\rightarrow$  numerical value of the area under ROC curve



Model-2



Model-2



1. Youden's J statistic

2. Distance to corner

$$J = \text{TPR} - \text{FPR} \quad \text{maximum}$$

$$d = \sqrt{(1 - \text{TPR})^2 + (\text{FPR})^2}$$

minimum distance

# Cross-validation

→ Evaluation only

Recap:

Data → Train → training  
 Data → Val → how well model is learning  
 Data → Test → final Test

Data → Train → less data  
 Data → Test

Solution: Instead of one fix split, we train multiple time, each with a different set.

Data (1000 rows)  
 → train (800) → test (0.2)

CV →  
 $k\text{-fold} = 5$   
 $\text{test\_ratio} = \frac{1}{k\text{-fold}}$   
 $= \frac{1}{5} = 20\% \rightarrow 160$   
 $\text{Train} = 800 - 160 = 640$

→ set-1 → [1,2,3,4,5,6,7,8] [9,10]  
 → set-2 → [1,2,3,4,5,6,9,10] [7,8]  
 → set-3 → [1,2,3,4,9,10,7,8] [5,6]  
 → set-4 → [3,4]  
 → set-5 → [1,2]

→ Each data becomes test data once

(0.01, prob.thr) performance  
 Set-1 → model (param-X) → 96%  
 Set-2 → 84%  
 Set-3 → 90%  
 Set-4 → 60%  
 Set-5 → 80%  
 average → ☐  
 (0.02, prob.thr)  
 → model (param-XL) → average → ☐

Data  
 → train  
 → mixed?  
 → hard? case -  
 → test  
 → mixed?  
 → simple? case -

→ evaluate → test data  
 → easy → hard

random Train Test  
 Model → 20 → 80 90  
 → 40 → 76 80  
 → 2 → 90 40  
 ↑  
 target

- goal-1 : evaluate model performance  $\rightarrow$  actual perf  
 goal-2 : when compared with models of different parameters, will help us do model selection

goal-1 : satisfied  $\rightarrow$   $\frac{\text{train}(\text{without } v)}{\text{all data}} \rightarrow \text{train-model} \rightarrow \text{model}(\text{final})$

$\downarrow$   
 goal-2 (Hyperparameter tuning)  $\rightarrow$  model selection

$\downarrow$   
 optimal value of parameters (learning rate, prob-threshold)

$\downarrow$   
 $\frac{\text{train}(\text{without } v)}{\text{all train}} \rightarrow \text{model} \rightarrow \text{final model}$

Train-Test  
 $\rightarrow$  stratified  
 (balance)

100 Blue marbles  $\rightarrow$  80 (Train)  $\rightarrow$  80 blue  
 100 Green marbles  $\rightarrow$  20 (Test)  $\rightarrow$  20 blue  
 20 green

$\rightarrow$  cross-validation split  $\rightarrow$  evaluation  $\rightarrow$  goal-1 (eval)  
 $\rightarrow$  goal-2 (hyper-parameter tuning)

stratified k fold  $\rightarrow$  stratified  
 $\rightarrow$  cross-validation  $\rightarrow$  Regression  
 $\downarrow$   
 Classification